2 **(a)** A speaker emits sound waves uniformly in all directions. Fig. 2.1 shows the variation with time t of the displacement x of an air molecule at a point Q that is 120 cm from the speaker.

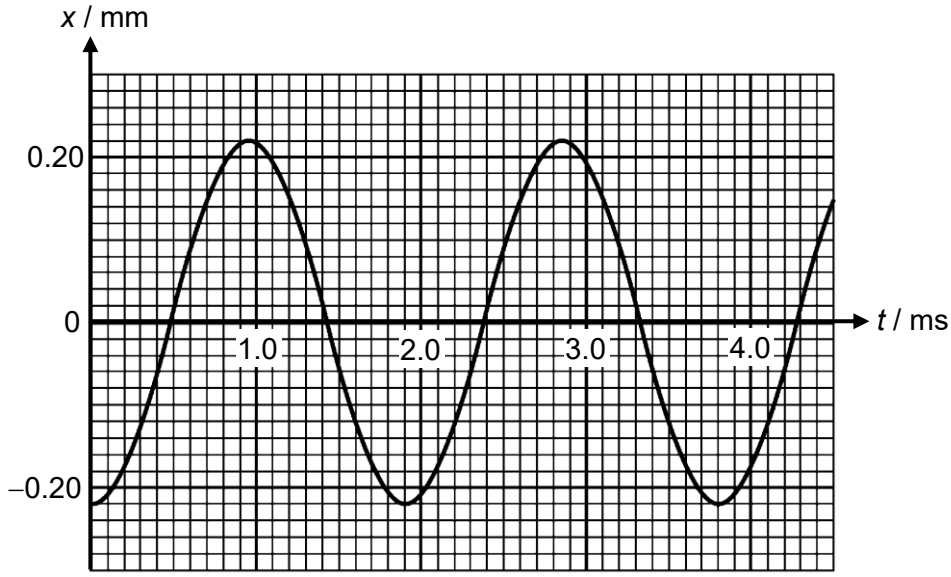

Fig. 2.1

(i) Use Fig. 2.1 to determine the

1. frequency f of the sound waves,

f = ... Hz [1]

2. uncertainty in f calculated in **(a)(i)1.** caused by reading the scale of the graph.

uncertainty in f = ... Hz [3]

(ii) Determine the next earliest time after 1.5 ms when the motion of the air molecule at Q has a phase difference of $\frac{4}{5}\pi$ compared to its phase at 1.5 ms.

time = ... ms [2]

(iii) If the power of the source is reduced to 0.25 of its initial value, calculate the distance from the speaker that will have the same intensity as that at point Q.

distance = ... cm [2]

(b) The wave arriving at point Q is progressive in nature. A stationary wave may be formed when two identical waves travelling in opposite directions superpose.

State the differences between the particles of a progressive wave and particles of a stationary wave in the following aspects:

(i) amplitude,

.....

.....

..... [1]

(ii) phase difference.

.....

.....

..... [1]